

Designing Large-Scale Wireless Sensor Networks for Urban Environmental Sensing

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ABSTRACT

Hardware and software advances have paved the way for large-scale environmental urban sensor networks, but there is no guidance on where to place a finite number of nodes and how to assess the network design. In this work, I develop and test two quality metrics for urban sensor networks—one that focuses on the impacts of urban form and one that acknowledges cities as social spaces. I then propose a data-driven algorithm grounded in urban planning theory for a new urban sensor network design. Through real-world deployments and simulations using open data, I will compare my proposed design and quality metrics to those used in prior sensor networks. This work will enable the deployment of environmental sensor networks that produce useful citywide data for numerous stakeholders despite the complexities of urban environments.

CCS CONCEPTS

• **Computer systems organization** → **Sensor networks**; • **Human-centered computing** → *Ubiquitous and mobile computing*; • **Applied computing** → Environmental sciences.

KEYWORDS

Sensor Networks, Urban Sensing, Environmental Sensing, Smart Cities

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1 PROBLEM STATEMENT

Urban residents are exposed to a variety of ambient environmental hazards, such as poor air quality, noise pollution, and extreme heat [11, 14]. These hazards contribute to increased health risks and mortality [14, 21, 28, 37], even with short-term exposure [18, 27], and will likely increase in the coming years as urban populations

grow [36] and climate change increases the occurrence of phenomena that contribute to these environmental harms [35]. Many researchers and policymakers rely on data from regulatory monitors to track pollutants, but these monitors are large and expensive, resulting in geographically sparse networks that cannot capture the known variability of pollutants at finer spatial resolutions [12, 26]. Furthermore, environmental hazards are often not equitably distributed throughout cities, making it difficult to prioritize mitigation efforts and alert residents of temporal or hyperlocal harms [14, 24, 25]. Advances in software and the lowering cost of hardware have paved the way for cities to use networks of low-cost sensing technologies to monitor environmental hazards and identify local hotspots. However, these networks are difficult to deploy because there is no guidance on how to design a sensor network in an urban space based on a city's budget—which affects the number of nodes available—and priorities. In particular, we lack information on 1) how to measure the quality of an environmental urban sensor network, and 2) where to place individual nodes to optimize for network quality. I address these challenges by developing two quality metrics for urban sensor networks—one that accommodates for the effect of urban form on environmental hazards and low-cost sensors, and the second that acknowledges the nature of cities as social, and not just physical spaces. I will then develop a method to design a sensor network that will provide users an option to optimize for different metrics, while exploring the tradeoffs made.

2 RELATED WORK

Few previously deployed urban sensor networks note how the networks were designed. Of those that do, some are placed randomly, either through completely random selection or through techniques such as stratified random sampling [6]. Networks that follow a deterministic design often use a grid-based layout [2, 19].

Area coverage, which is a measure of how much physical area is being monitored by a sensor network [31], is often used as the quality metric for sensor networks, but it is inadequate for urban environmental sensing for two main reasons. First, area coverage generally considers spaces as two-dimensional planes, but cities are inherently three-dimensional spaces, with structures and other urban forms that must be considered. For example, high-rise buildings can affect wireless signal, impacting where sensor nodes are placed due to connectivity issues [3], and urban form such as building height, road density, and amount of greenery greatly impact environmental factors including air quality and temperature [11, 34]. Second, cities are social and political, not just physical spaces, and those features must be considered when measuring the quality of an urban sensor network, as nodes must often be placed based on social and/or political constraints [7, 30].

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Some researchers have developed alternative quality metrics that consider the social implications of urban sensor networks. One such approach, "citizen-centric objectives," [32] considers a range of factors to optimize sensor placement. These factors include measuring vulnerable populations' exposure to air pollution by placing sensors near schools and senior homes, maximizing public satisfaction by minimizing the distance to sensors for most people, and monitoring traffic emissions by optimizing sensor placement along specific road segments based on traffic patterns [33]. Another metric, called "cost-constrained multi-resolution dynamic mode decomposition", factors in the multi-scale variability of air pollution and two socioeconomic indicators—race and income—to determine optimal sensor placement [13]. Similarly, researchers have examined "sensor deserts," evaluating sensor networks based on the straight-line distance between a sensor and neighborhoods with certain income levels or racial makeup [30]. Although these approaches improve upon the limited model of area coverage, they tend to focus on demographic measures of social vulnerability rather than the specific hazards experienced by communities [25]. Furthermore, these approaches assume a large, binary sensing region for low-cost sensors, where entire neighborhoods are often "sensed" when they have just one sensor. This model does not account for the effect of urban form on the sensors, and thus overstates their coverage.

3 METHODOLOGY

3.1 Identify Data Users and Their Priorities

In collaboration with the Urban Innovation Initiative at Microsoft Research, I helped deploy and maintain a large-scale air quality sensor network in Chicago [6]. Through the project, we identified the primary data users of urban environmental sensor networks and learned about users' priorities and data needs—both which affect where nodes should be placed. We noted that there are the three key data consumers for environmental sensor networks—cities, researchers, and community members—and that all three user groups have shared priorities in achieving better health outcomes and revealing emissions sources as goals of urban environmental sensor networks [7]. Furthermore, we found that city and community stakeholders have shared motivations around education, engagement, and empowerment. I use these insights to help guide the remainder of my thesis work around developing new sensor network quality metrics and a design strategy.

3.2 Develop Quality Metric that Incorporates Urban Form

To satisfy researchers and help provide more accurate data for public health and emissions sources, I am working on a quality metric around the certainty of the sensor readings. For this metric, I consider the effect of urban form on the environmental hazards measured. Prior research indicates that air quality and noise differ based on the street type [23, 25], which is classified using the ratio of road width to building height [25], resulting in three different street types—high, medium, or low flushing—as shown in Fig. 1.

To calculate the certainty, I incorporate the inherent error that comes with using low-cost sensors [16] and the added uncertainty as we move further away from a sensor. The certainty is thus never equal to 1.0 (100%), and is always decreasing with the distance from

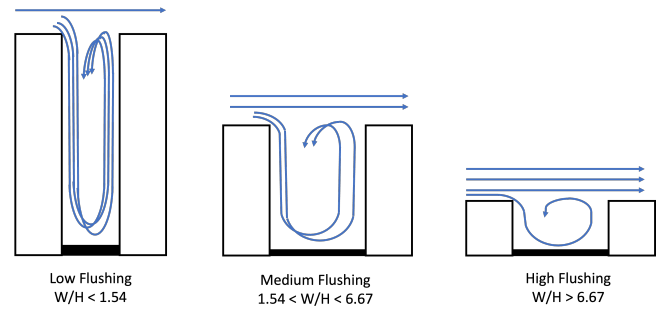


Figure 1: Illustration of different types of urban streets, based on the road width (W) to building height (H) ratio, and its effect on the motion of airborne pollutants. Similar effects have been found with urban form and noise levels as well [23].

the sensor. For the equation, I use a Gaussian function based on the assumed Gaussian spatial distribution of the underlying environmental phenomena [5, 8, 15]. I then calculate the certainty in a step-wise manner block by block. I use this approach because: 1) it fits in line with findings around air quality, heat, noise, which can all differ from block to block—sometimes even within the same block. Yet most blocks have a singular urban design that persists for the length of that block [25], 2) it is easy to explain to various stakeholders (i.e. city officials, community groups, residents, researchers from various fields), as it fits with urban design and urban planning and the general understanding of cities, 3) although it may not be particularly precise for within-block variations, it provides a level of granularity beyond many of the existing approaches today, and importantly is at a level to enact change.

For a block that has a sensor, the certainty is equal to i , the initial certainty based on the type of sensor and street type. Moving in a single direction from the sensor, I then calculate the certainty block by block. For blocks that are all the same type (high, medium, or low flushing), the certainty is calculated with this equation:

$$f(n) = i * e^{-\frac{n^2}{2c^2}} \quad (1)$$

where n is the number of blocks away from the sensor and c is the standard deviation of certainty based on the street type. In the current iteration, I use $c = 3$ for low flushing streets, $c = 2$ for medium flushing streets, and $c = 1$ for high flushing streets. This can be viewed as saying that we have the most certainty within 3, 2, and 1 block, respectively, for these street types.

If a block type differs from the prior one, I use the following equation, which is visualized for a block change in Fig. 2:

$$f(n) = f(n-1) * e^{-\frac{x-n^2}{2c^2}} \quad (2)$$

I have completed the design of this algorithm, as shown in Algorithm 1, and the equation, as shown in Equations (1) and (2), for the case of calculating the certainty of each block moving in either direction along the same street from a single sensor. The output of this algorithm on a small test area of Boston is shown in the top panel of Fig. 3. As shown in the lower panel of Fig. 3, almost all of the streets in this neighborhood are low flushing, and the street

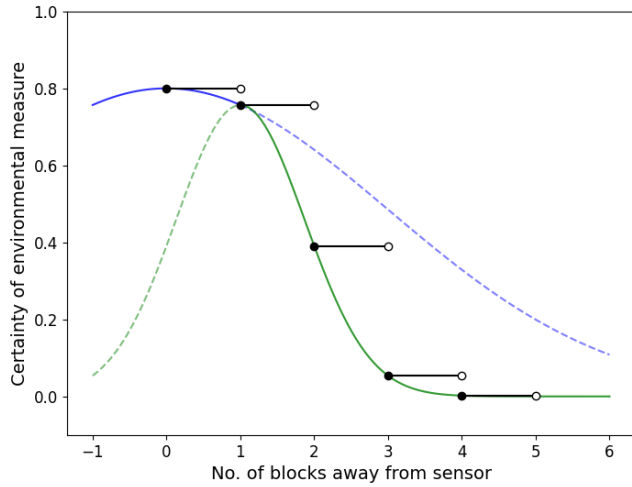


Figure 2: This graph shows the output of the stepwise Gaussian-based function for a potential street. The street changes from low to high flushing one block from the sensor, thus c changes from 3 to 1. The dotted lines represent the continuation of each separate function on either side of the intersection and the solid lines show the function that is used given the change in street type.

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i ← initialcertainty;
prior_type ← initialblocktype;
while (certainty > 0) and (street continues) do
  if block_type ≠ prior_type then
    do equation 1;
  else
    do equation 2;
  end
  prior_type = block_type
end

```

Algorithm 1: Basic algorithm to set the certainty on a single street with a single sensor based on equations 1 and 2

types do not change from one to another along the same street. From this small case study, and prior research [17], it is clear that streets may be more homogeneous in certain neighborhoods, thus it may be useful to consider other data beyond street type, such as density of restaurants or traffic patterns.

The next steps are to determine how to calculate certainty with multiple sensors on the same street, how to calculate certainty around corners, and which other data to include for more accurate estimations.

3.3 Develop Quality Metric to Address the Social Nature of Cities

Cities are more than just physical spaces, and stakeholders of environmental sensor networks include more than only researchers. Thus, the quality of an environmental sensor network must also

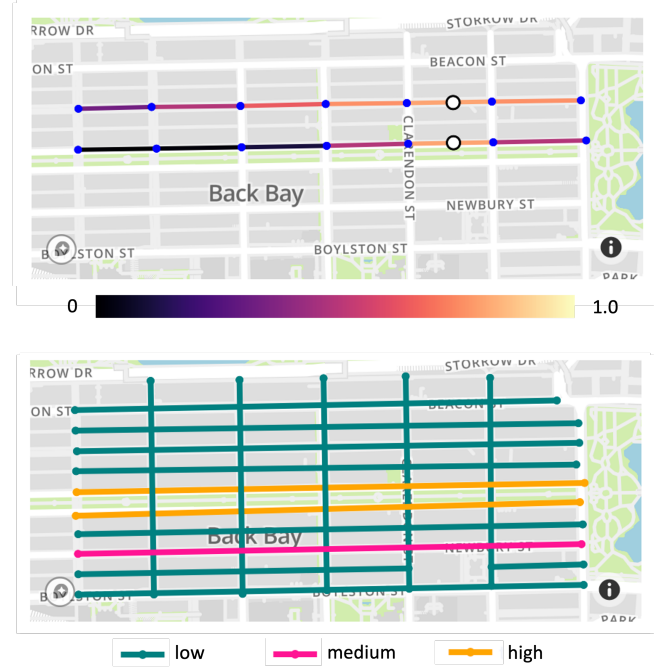


Figure 3: These images show a small area of Boston with a mix of low, medium, and high flushing streets. The top image shows the output of the algorithm, with two potential sensor placements (white circles outlined in black). The initial certainty is set to 0.8 for both sensors, and the top street is low flushing ($c = 3$) while the bottom street is high flushing ($c = 1$). Thus we have more certainty moving further away on the top street. In the bottom image, we show the flushing types for this neighborhood. The entire neighborhood is quite homogeneous in terms of street type, raising questions of when streets will change and how to test that.

be measured by how well the network achieves social priorities, such as equity. Equity, which has often been overlooked in urban sensor network and smart city research [22], is essential in urban sensor network design 1) to ensure that the deployment of sensor networks will not further the digital divide that exists within and between most cities [22] and 2) because most urban environmental hazards are inequitably distributed [14, 24, 25], so nodes must be strategically placed to help cities identify problem areas and develop appropriate mitigation strategies.

To this end, I will develop a second quality metric that measures how equitably an urban sensor network is distributed throughout the city. Although some researchers have begun advances in this direction [13, 30, 32, 33], these prior efforts focus solely on racial and income demographics, which do not always correlate with areas of higher environmental hazards [25]. In addition, relying on racial demographics represents a Western view, and thus will not generalize to cities that are more racially homogeneous, including many of those in the Global South. I will thus explore alternative measures of equity to incorporate in this metric, including equal representation of street and neighborhood types, prioritized coverage of

neighborhoods or locations with higher population of children or seniors, who are more susceptible to environmental harms [24], and equity based on certainty of the sensor network, as defined by my prior quality metric. Because I have not fully dug into this portion of my thesis yet, I expect that I will devise alternative metrics of equity, and perhaps other social metrics worth considering.

3.4 Design Strategy to Achieve Quality Metrics

After completing the development of the two new quality metrics, I will design a new strategy to determine where to place the sensor nodes to achieve those quality metrics. I plan to devise the strategy based on the theoretical grounding of public facility location, which determines where to place stationary public services, such as libraries, and mobile services, such as ambulances, relative to the population density across a city [20]. Rather than focusing solely on population density, I will explore developing an algorithm that focuses on other measures. For example, I will consider environmental risk factors, such as age and pre-existing health conditions [24]. One potential objective function for this network design algorithm is thus to maximize the number of high-risk exposure hours monitored, which will amplify the shared priority around health exposures among all three stakeholder groups [7].

In addition, to accommodate for the social equity metric, I plan to utilize work on socially fair facility location [10] to develop an objective function that balances the needs of different populations and the various stakeholder groups. My goal is to devise an algorithm that provides solutions to optimize for both of the quality metrics I design, allowing users to weigh the metrics differently and see how the network design changes with the different weights. Because the deployment of limited sensing resources in urban spaces inevitably requires trade-offs, and because each city is unique and will likely have its own priorities, it is essential to allow cities to explore different designs rather than provide a prescriptive option.

4 EVALUATION

4.1 Determine Data Structure for Cities

Testing the different aspects of my thesis requires creating a digital format of a city to run analyses before or in lieu of real-world deployments. Using inspiration from urban transport and urban planning research [1, 4], as well as recent work on urban digital twins [9], I will represent urban spaces using a graph data structure to capture both the topology and geography of city streets. I am still working to determine the best method to capture the necessary data, which includes intersections, blocks, latitude, and longitude to know which blocks intersect and how. Currently I plan to represent each block as an edge and each intersection connecting blocks as a vertex in an undirected graph, but future iterations could incorporate data such as wind or traffic direction in a directed graph.

4.2 Field Work

To evaluate the equation I use for estimating the certainty of values recorded in a sensor network, I am designing a field experiment using low-cost air quality, noise, and temperature sensors in three small areas of Boston, one of which is shown in Fig. 3. By comparing the sensor readings on different blocks that have a mix of low, medium, and high flushing street types, I will gather data to verify

the validity of my equations and parameters. Additionally, I will use previously collected air quality and temperature data from the sensor network in Chicago and combine those data with road and building information to further validate the equations.

4.3 Real-World Deployments

Both of the quality metrics and the network design strategy I develop will be evaluated using real-world urban sensor deployments in Chicago and Boston. I choose to work with these two cities because they have adequate open data, have sensor networks previously or currently deployed, and I am familiar with their geography and demographics. First, I will use the large-scale Microsoft Eclipse network in Chicago [6] and other existing sensor networks in Chicago and Boston, such as the PurpleAir network [29], to examine how the new quality metrics I develop compare to the classic area coverage measurement. I will then work with local collaborators in the Boston area to test my network design in two upcoming NSF-funded environmental sensing networks. Specifically, I will evaluate how well my proposed design algorithm fulfills each metric compared to existing network designs such as stratified random sampling, and if my proposed network design allows for the same scientific analyses as existing designs. Furthermore, I will examine whether my proposed design allows for fewer sensors to be used or offers insights into new research agenda.

4.4 Simulations

To ensure that my thesis work can generalize to cities beyond Boston and Chicago, I will determine how easily my proposed network designs can be deployed to other cities based on the simple availability of open data or geographic information such as maps. I expect that other cities with open data will be easy to incorporate into my graph data structure, and then simulate the placement of sensors to evaluate my quality metrics and design method. For cities that lack open data, I will explore the possibility of using maps or satellite images to create graph structures then run the analyses.

5 CONTRIBUTIONS

This work will provide numerous contributions to researchers and practitioners who wish to deploy sensor networks to gather urban environmental data. It will provide the means to measure the quality of an urban sensor network, allowing those who deploy networks to gauge how well their network will perform in terms of both scientific data collection and social equity. By grounding these metrics in the use of urban data and learnings from prior deployments, I will provide a framework for those who work with sensor networks in other settings to develop their own specified quality metrics. Through my network design algorithm, I will offer a generalizable solution for cities and researchers to deploy sensor networks, ensuring that networks can capture representative data throughout the city while also achieving goals of equity and scientific rigor. This work will aid in increasing the number of sensor networks deployed around the world, helping to bring the environmental smart city vision from a dream to a reality.

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